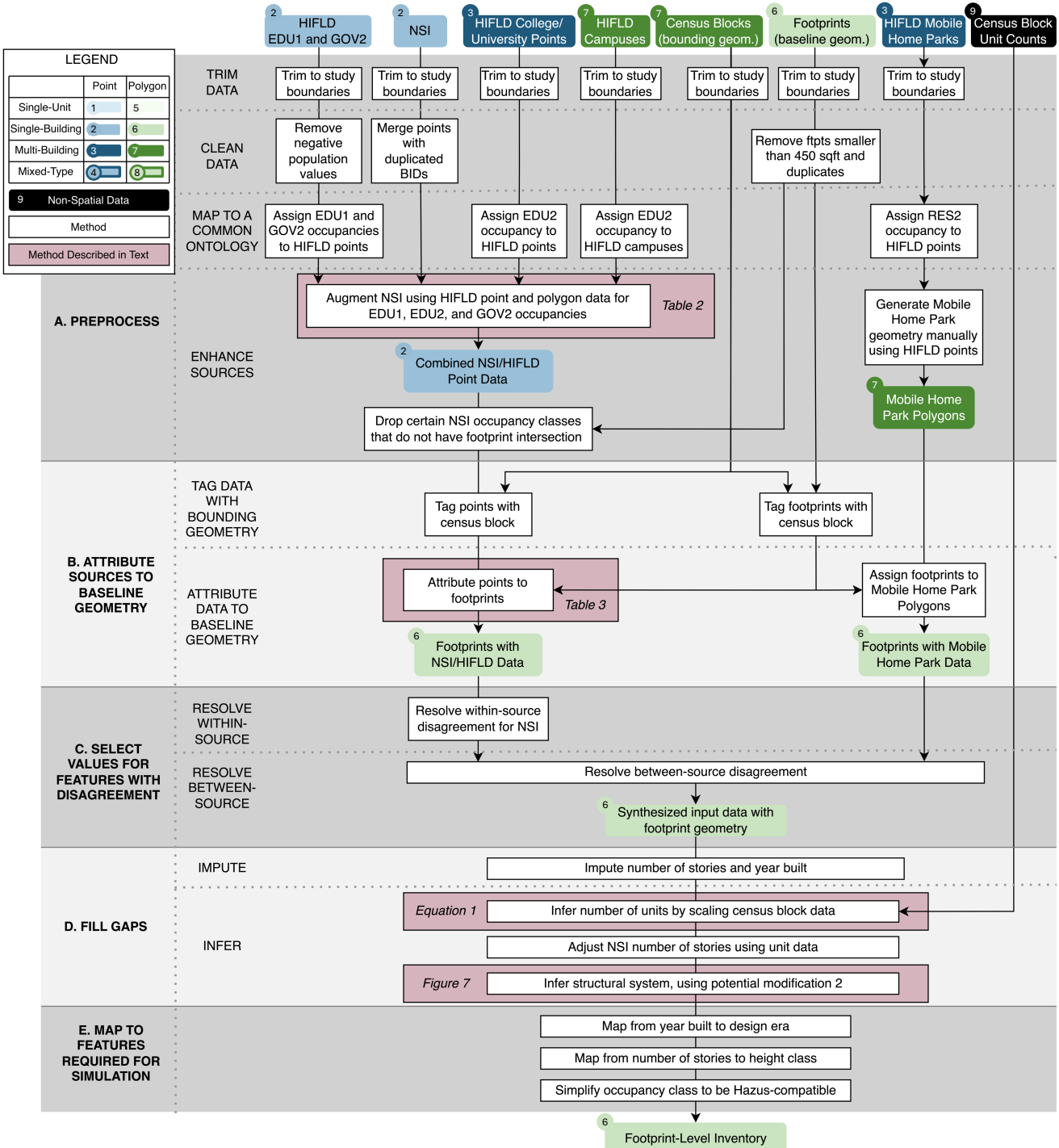
LEGEND
Point
Polygon
Single-Unit
Single-Building
Multi-Building
Mixed-Type
Non-Spatial Data
Method
Method Described in Text
2 HIFLD EDU1 and GOV2
2 NSI
3 HIFLD College/ University Points
7 HIFLD Campuses
7 Census Blocks (bounding geom.)
6 Footprints (baseline geom.)
3 HIFLD Mobile Home Parks
9 Census Block Unit Counts
A. PREPROCESS
TRIM DATA
Trim to study boundaries
CLEAN DATA
Remove negative population values
Merge points with duplicated BIDs
Remove ftpts smaller than 450 sqft and duplicates
MAP TO A COMMON ONTOLOGY
Assign EDU1 and GOV2 occupancies to HIFLD points
Assign EDU2 occupancy to HIFLD points
Assign EDU2 occupancy to HIFLD campuses
Assign RES2 occupancy to HIFLD points
ENHANCE SOURCES
Augment NSI using HIFLD point and polygon data for EDU1, EDU2, and GOV2 occupancies
Table 2
Combined NSI/HIFLD Point Data
Generate Mobile Home Park geometry manually using HIFLD points
Mobile Home Park Polygons
Drop certain NSI occupancy classes that do not have footprint intersection
B. ATTRIBUTE SOURCES TO BASELINE GEOMETRY
TAG DATA WITH BOUNDING GEOMETRY
Tag points with census block
Tag footprints with census block
ATTRIBUTE DATA TO BASELINE GEOMETRY
Attribute points to footprints
Table 3
Assign footprints to Mobile Home Park Polygons
Footprints with NSI/HIFLD Data
Footprints with Mobile Home Park Data
C. SELECT VALUES FOR FEATURES WITH DISAGREEMENT
RESOLVE WITHIN-SOURCE
Resolve within-source disagreement for NSI
RESOLVE BETWEEN-SOURCE
Resolve between-source disagreement
Synthesized input data with footprint geometry
D. FILL GAPS
IMPUTE
Impute number of stories and year built
INFER
Equation 1
Infer number of units by scaling census block data
Adjust NSI number of stories using unit data
Figure 7
Infer structural system, using potential modification 2
E. MAP TO FEATURES REQUIRED FOR SIMULATION
Map from year built to design era
Map from number of stories to height class
Simplify occupancy class to be Hazus-compatible
Footprint-Level Inventory